ARCANGELO ALBERICO

Software Engineer | Data Engineer | AI Engineer

Age: 30 Bologna albericoarcangelo@hotmail.it arclk.dev arclk arclk



EXPERIENCE

Software Engineer & Data Engineer

Iconsulting S.p.A.

Nov 2021 – Ongoing Bologna, Italy

- Development of an agent-based system for natural language interaction with business intelligence dashboards. Built on an open-source conversational AI platform, featuring visual components using RAG, query translation for reporting, and a semantic BI model with QnA-style visuals. Includes an advanced query agent leveraging knowledge base semantics for automatic analytical query generation.
- Design and develop full-stack web applications using FastAPI, Python, SQL, and Angular for analyzing and classifying sales of manufacturing products based on historical data, enabling data-driven decision-making. Deployed the web application with a CI/CD on AWS Elastic Beanstalk, ensuring secure and scalable hosting.
- Develop a middleware system for seamless communication and data synchronization between two systems, utilizing AWS SQS for message queuing, AWS Lambda functions for event-driven computation, and AWS DynamoDB for storing and enriching data based on the middleware's information. Used Terraform for provisioning and managing the AWS infrastructure as code, ensuring consistent and reproducible deployments.

Internship

Engineering Ingegneria Informatica S.p.A

April 2017 – Sept 2017 Napoli, Italy

Develop Smartroad, a web application using Angular as client-side and Spring as server-side frameworks, which calculates incidentality indices before and after a maintenance work on a road section.

EDUCATION

M.Sc. in Artificial Intelligence

Alma Mater Studiorum – Università di Bologna

Oct 2019 – Oct 2021

Thesis title: A Neurosymbolic Framework for Markov Logic Networks
Mark: 110/110 cum laude

Artificial Intelligence: Algorithms and Applications

Nankai University, China

July 2021

Online Summer School Program focused on Machine Vision and Medical Robotics

B.Sc. in Computer Engineering

Università degli Studi di Salerno

Sept 2014 – Sept 2017

Thesis title: Sviluppo di una Web Application per il calcolo degli indici di incidentalità di tratte stradali
Mark: 110/110 cum laude

High School Diploma

Liceo Scientifico Federico Quercia, Marcianise (CE)

Sept 2009 – July 2014

Mark: 100/100

CERTIFICATIONS

AWS Solutions Architect – Professional
AWS Certified Developer - Associate
Claude Certified Developer - Foundations

PROJECTS

I realized several projects during my university carreer or free time available on my Github. Among them:

- PyTorch implementation of a neurosymbolic framework that integrates neural networks with Markov Logic Networks.
- A NetLogo implementation of the SEQIJR epidemiological model for studying the evolution of the COVID-19 epidemic.
- An object detection systems to identify single and multiple instances of products in given scene images.
- Text mining analysis on clickbait titles, aiming to identify the causes and patterns that make a title a clickbait.
- Solving the Perfect Weighted Packing (PWP) problem, which involves packing a set of rectangles into a strip without overlapping.

PERSONAL SKILLS

Team work, perseverance, problem solving, communication, critical thinking

TECHNICAL SKILLS

Languages: Python JavaScript R C Java C# Scala Prolog SQL HTML PHP CSS LaTeX
Artificial Intelligence: Pytorch Tensorflow LLM RAG Agents LangChain
Web Development: FastAPI Angular React .NET
Cloud: AWS Services Terraform

LANGUAGES

Italian - Mother tongue
English - IELTS 7.0 (C1)